# Corrected and conduct of chapter 4

**Use as a teacher.** The following answers relate to the outstanding tasks in the course. first clarify the approach; provide graduated help before the solution. estimates to be calibrated in class.

## Diagnosis and Remediation

1. $\frac{8-2}{3-1}=3$. *Action:* Represent horizontal and vertical increments.

2. $4h+h^2$. *Action:* Develop before simplifying.

3. $4+h$. *Action:* Factorize h to the numerator before division.

4. 0 and 0,1. *Action:* Check by square elevation.

5. 3 and 2. *Action:* Represent distance at 0.

6. $y=2x+3$. *Action:* Replace the coordinates to find the original order.

## Research activity

The slopes are 2 + $h$ ; limit 2. The line passes through (1; 1), therefore $y=2x-1$. The drawing motivates, the calculation justifies.

## Immediate applications

1. $f(2+h)-f(2)=7h+h^2$ ; rate 7 + $h$ for $h\neq 0$.

2. The rate is 4 + 3 $h$, so the derivative number is 4.

3. $y=5-2(x-3)=-2x+11$; approximation 5 $-2(0{,}04)=4{,}92$.

4. The rate in 2 is $\frac{|h|}{h}$, with the two opposite limit values. For the root in 0, the rate $\frac{1}{\sqrt{h}}$ for $h>0$ exceeds any bound when h approaches 0: no actual derivative number.

## Exercises: approach and results

**Exercise 4.1 · Two secantes.** Pentes 3 and 2.1; derived number 2. The second is closer to the tangent slope; this is an exact rate calculation.

**Exercise 4.2 · A tangent.** Rate 4 $a+2h=-4+2h$, therefore $f'(-1)=-4$. $f(-1)$ = 3 ; tangent $y=3-4(x+1)=-4x-1$.

**Fiscal year 4.3 · Marginal cost.** $C(10)=400$, $C(11)$ = 441, increment 41 for one unit. The rate around 10 is 40 + $h$, thus derivative 40 : approximation of the increment for a small variation, not equal for an entire unit.

**Exercise 4.4 · Read a tangent.** In the course figure: points (1;0) and (2;4), slope 4. the new situation, $f(2)$ = 3, slope $-\frac{1}{2}$. Approximation 3 $-0{,}5\times 0{,}1=2{,}95$. Ordered from B is not necessarily $f(4)$.

**Exercise 4.5 · Control an approximation.** $1+2(-0{,}02)=0{,}96$. Right 0 ,9604; exact less approach 0 $,0004>0$. The term omitted is $h^2$ : default approximation.

**Exercise 4.6 · Tangente and intersection.** The tangent has as an equation $y=1+3(x-1)=3x-2$. The development of the product gives well $x^3-3x$ +2. The apses of the intersections are therefore 1 and $-2$, by the zero-product rule. The points are (1; 1) and ( $-2;-$There are two commonalities: This is enough to refute the general statement.

**Exercise 4.7 · Numerical secants.** List of rates 2+ $h$ : 2.1; 2.01; 2.001 and 1.9; 1.99; 1.999. The algebraic calculation gives the real 2. At very small hours, the floating calculation can lose precision.

**Exercise 4.8 · Deferred review.** For $h\neq 0,-1$, the rate is $-\dfrac{1}{1+h}$ ; it tends towards $-1$. Approximation 1 $-0{,}02=0{,}98$. The exact value $\dfrac{50}{51}$ allows for control.

**CQFD, Exercise 17 p. 127, all questions.** The apses of A, B and C are 0, 1 and 2. tangents give respectively slopes 2, 0 and $-2$. Thus $f\prime(0)=2,\ f\prime(1)=0,\ f\prime(2)=-2$. The axis units shall be read on the grid.

**CQFD, Exercise 19 p. 127, Questions 1 and 2.** Point A reads (1; 4), therefore $f(1)=4$. The tangent passes through (0; 1) and (1; 4): slope 3. $f\prime(1) = 3$ and tangent a for equation $y=4+3(x-1)=3x+1$.

## MCQ: answers and reasons

**MCQ 1.**

1. B. $\dfrac{4-1}{2-1}=3$.

2. A. $4+2(x-1)$.

3. D. This is a value in one point.

4. C. Rate 6 + $h$.

5. B. Affine tangent function.

**MCQ 2.**

1. C. Rates $\dfrac{1}{\sqrt{h}}$ Not finished.

2. A. $|h|=-h$.

3. D. Constant order.

4. B. A change in position divided by a change in time is expressed in m s $^{-1}$.

5. C. Accurate equality for an affine, approximation in general.

## True or false

1. False: $x^3$ and tangent 3 $x-2$ in 1 also meet in $-2$.

2. False: Rates on both sides are valid $-1$ and 1.

3. True: the rate is exactly m for any non-zero h.

**Differentiation.** For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.